

How Does a Game Place Sound Around You?

國中 7 年級

第 3 週

預計 94 分鐘

家長只需要做一件事

孩子完成學生教材後，再把這份答案交給他自行核對。您不需要講課，也不需要逐題追問。

本週學習重點：本週改用遊戲 spatial audio，閱讀難度再微升：孩子要同時追蹤 distance、direction、obstruction 三種關係，並學新的時間/位置介系詞。上一週命令句不重教，只保留一個完整步驟產出作複習。

觀察重點：

- 是否能分清 distance、direction/orientation、obstruction，而不是看到聲音變化就猜單一原因。
- 是否能正確使用 at/on/in 與 behind/in front of 等位置片語。
- 跨段題是否會先找兩段以上證據再下結論。
- 三步驟命令句是否能自己完整寫出，而不是只會辨認正確句型。

完成度確認：完成 10 個核心字、約 390 字閱讀、三關係 organizer、五階段練習與三題分日作業。

預計總時間：94 分鐘

本週調整說明：依回饋繼續科技主題但換成遊戲 3D 音訊，避開連續重複 AI stem separation。；閱讀再升一小階，從兩工具比較改成三種空間關係整合。；主要文法前進到 g7-time-place-prepositions；imperatives 只做 evidence-backed spaced review。

本週內容參考：Apple Developer Documentation、Roblox Creator Hub、Meta Horizon OS Developers

答案與解析

G1 — The bell is ____ the table in the test scene.

答案：A

也可接受：on

簡短理由：TEST SCENE 明確寫 A bell is on a table。物體放在桌面上用 on。

常見誤區：選 in 會變成鈴鐺在桌子裡面。

G2 — In paragraph 1, what is the listener?

答案：B

也可接受：The point from which the player hears the scene

簡短理由：第一段直接定義 The listener is the point from which the player hears the scene，因此 B。

常見誤區：不要把 listener 和 sound source 對調。

G3 — The bell stays in the same position, but the player walks nearer to it. Which relations...

答案：C

也可接受：The distance between source and listener

簡短理由：bell 的 position 固定，玩家走近，所以 source 與 listener 之間的 distance 變短。

常見誤區：題目沒有新增牆，也沒有改測試時間。

I1 — Complete the organizer for distance, direction, and obstruction. For each relationship,...

答案：Sample: Distance—walk nearer to the same source; a closer source can play more loudly. Direction/orientation—turn around while the source stays fixed; the source can remain in its virtual position relative to the scene. Obstruction—move behind a wall while the source stays fixed; PHASE can use the obstruction and may lower the sound volume.

也可接受：Any three-row organizer that correctly pairs each relationship with one reading-supported action and effect/response.

簡短理由：三列都要同時寫『改變的關係』與『正文支持的反應』。distance 對應遠近與音量；direction/orientation 對應轉向；obstruction 對應牆等遮蔽物。

常見誤區：只抄三個名詞而沒有 action/effect，沒有完成 organizer 的整合要求。

I2 — A creature stays in the same position while the player turns around. What changed most...

答案：B

也可接受：The listener's orientation relative to the source

簡短理由：creature 沒移動，玩家只是轉身。第六段明確說 turning around changes the listener's orientation relative to the source。

常見誤區：轉身不等於走近，所以不能推成 distance 變短。

I3 — In paragraph 4, obstruction most nearly means ____.

答案：C

也可接受：something between positions that can block or affect the sound

簡短理由：第四段把 wall 放在 player 與聲音之間，並說 system may use an obstruction as information，因此 obstruction 是會擋在中間、影響聲音關係的東西。

常見誤區：不要因為同段有 volume 就把 obstruction 當成音量設定。

C1 — What is the main idea of the reading?

答案：B

也可接受：Spatial audio can use source-listener relationships such as distance, direction, and obstruction to change what a player hears.

簡短理由：全文依序處理 source/listener position、distance、direction、environment/obstruction，最後要求指出哪個關係改變。B 才能涵蓋全文。

常見誤區：A、D 都把文章的局部例子過度擴大成所有遊戲或唯一原因。

C2 — What do the Apple and Roblox examples in paragraph 2 both support?

答案：A

也可接受：Distance between listener and source can influence the volume of positional audio.

簡短理由：第二段分別用 Apple PHASE 與 Roblox positional audio 支持同一關係：listener-source distance 會影響音量。文章沒有說所有遠聲都完全消失。

常見誤區：看到兩個品牌時要找共同支持的命題，不要把其中一家的系統名稱套到所有遊戲。

C3 — A player walks nearer to a creature and also moves behind a wall. The sound changes. Wh...

答案：C

也可接受：Because both distance and obstruction changed, so the test does not isolate one relationship.

簡短理由：玩家同時走近又移到牆後，distance 與 obstruction 都變了。只看最後聲音不同，無法把差異唯一歸因給 distance。

常見誤區：能找到一個可能原因不代表已排除另一個同時改變的原因。

P1 — Write exactly two complete sentences describing the TEST SCENE. Use at least three plac...

答案：At 8:00, the player is in the training room. The bell is on the table in front of the player.

也可接受：Any exactly two complete sentences that accurately describe the TEST SCENE and use at least three correct time/place prepositions or position phrases.

簡短理由：示範答案剛好兩句，正確使用 at 8:00、in the training room、on the table、in front of。

常見誤區：常見錯誤是用 in 8:00 或 in the table。

P2 — Write exactly two complete sentences explaining one audio change with because. Name the...

答案：The bell may sound louder because the listener moved nearer to the same source. The change is linked to distance because the bell stayed in the same position.

也可接受：Any exactly two complete sentences that name one reading-supported relationship, state a supported effect, and identify one controlled condition.

簡短理由：兩句都保留 same source / same position 的控制條件，再把 nearer 連到 distance，因此沒有把其他關係偷偷混進來。

常見誤區：如果同時寫又走近又躲到牆後，就不能再說只有 distance 造成變化。

R1 — The test begins ____ 8:00 ____ the training room.

答案：C

也可接受：at / in

簡短理由：8:00 是精確時間點，用 at；training room 是人在其中的空間，用 in。

常見誤區：on 常用在星期/日期或表面，不用在 8:00。

R2 — Write exactly three imperative sentences for a fair distance test. Keep the source in t...

答案：Sample: Keep the sound source in the same position. Walk farther from the source.

Compare what you hear with the first position.

也可接受：Any exactly three grammatical imperative sentences that keep source position fixed and change only listener distance.

簡短理由：這是上週 imperatives 的 spaced review。三句都以原形動詞開頭，而且只改 listener distance，不同時改 source 或 obstruction。

常見誤區：若同時 Move the source and walk away，就違反題目要求的單一改變。

H1 — The game event is ____ Friday ____ 7:30 p.m.

答案：B

也可接受：on / at

簡短理由：Friday 是星期，用 on；7:30 p.m. 是精確時間，用 at。

常見誤區：不要把日期/星期的 on 和鐘點的 at 對調。

H2 — A sound source moves from in front of the player to behind the player. The distance sta...

答案：D

也可接受：Direction relative to the listener

簡短理由：題目明確固定 distance 且沒有 wall，只把 source 從 in front 移到 behind，因此清楚改變的是 direction。

常見誤區：front/behind 是方向關係，不代表距離一定改變。

H3 — Write exactly three complete sentences explaining why a test that changes both distance...

答案：The test changes both distance and obstruction at the same time. The reading says distance can affect volume, and it also explains that a wall can act as an obstruction that changes the sound. Because two relationships changed, the result cannot show which one alone caused the audio difference.

也可接受：Any exactly three complete sentences that name both changed relationships, use at least one specific reading idea, and explain why single-cause attribution is unsupported.

簡短理由：示範答案剛好三句，先指出兩個變項，再引用 distance 與 wall obstruction 的正文概念，最後說明為何不能唯一歸因。

常見誤區：只說『不公平』但沒指出 distance 和 obstruction 兩者都變，解釋仍不完整。

遇到不懂時

請孩子先找出錯在哪裡；仍不懂時，再拍下題目請 AI 解釋原因並另出一題。詳細方法請看網站的學習指南。